

Partial Positive Results for Complementing Bilinear Maps in Two-Lipschitz Spaces

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Source Question

Mandal's paper [1] asks in Question 1:

Is $\text{Blin}(X, Y; E)$ complemented in $\text{BLip}_0(X, Y; E)$?

The paper proves a norm-one positive answer when the codomain E is a dual Banach space and states that the general case remains open. This note records two partial extensions.

Definitions

For real normed spaces X, Y and a Banach space E , a map $T : X \times Y \rightarrow E$ belongs to $\text{BLip}_0(X, Y; E)$ if it vanishes on the coordinate axes and

$$\|T(x, y) - T(x', y) - T(x, y') + T(x', y')\| \leq C \|x - x'\| \|y - y'\|$$

for all $x, x' \in X$ and $y, y' \in Y$. The least such C is denoted $\text{BLip}(T)$. The closed subspace $\text{Blin}(X, Y; E)$ consists of bounded bilinear maps, with the usual bilinear norm.

Idea of the Proof

Mandal's projection is built from a vector-valued invariant mean. The dual-space hypothesis is only used to guarantee that the scalar invariant mean applied coordinatewise lands back in E , rather than merely in E^{**} . Thus the same formula works whenever the canonical copy of E is complemented in E^{**} .

There is also a different operator-theoretic route. The source paper recalls the two-Lipschitz linearization

$$\text{BLip}_0(X, Y; E) \cong L(\mathcal{F}(X) \widehat{\otimes}_\pi \mathcal{F}(Y), E).$$

Under this identification, bilinear maps are precisely the operators that factor through the tensor of the barycenter maps

$$\beta_X \otimes \beta_Y : \mathcal{F}(X) \widehat{\otimes}_\pi \mathcal{F}(Y) \longrightarrow X \widehat{\otimes}_\pi Y.$$

If those barycenter quotient maps have bounded linear right inverses, the tensor quotient splits and hence precomposition gives a projection for every codomain E .

Theorem 1 (codomains complemented in the bidual). *Let X, Y be real normed spaces and let E be a Banach space. Suppose that the canonical copy $\kappa_E(E)$ is complemented in E^{**} by a projection $R : E^{**} \rightarrow \kappa_E(E)$ with $\|R\| \leq C$. Then $\text{Blin}(X, Y; E)$ is complemented in $\text{BLip}_0(X, Y; E)$ by a projection of norm at most C .*

Proof. Let $G = X \times Y$ be the additive amenable group. Choose a scalar invariant mean m on $\ell_\infty(G)$. For $f \in \ell_\infty(G, E)$ define $\widetilde{M}f \in E^{**}$ by

$$\langle \widetilde{M}f, e^* \rangle = m(\langle e^*, f(g) \rangle)_{g \in G} \quad (e^* \in E^*).$$

Then $\widetilde{M}$ has norm one, is translation invariant, and sends the constant function $g \mapsto e$ to $\kappa_E(e)$. Set

$$M = \kappa_E^{-1} R \widetilde{M} : \ell_\infty(G, E) \longrightarrow E.$$

This is an E -valued translation-invariant mean with $\|M\| \leq C$ and $M(\widehat{e}) = e$ for constant functions.

For $T \in \text{BLip}_0(X, Y; E)$ and $w \in X, z \in Y$, put

$$\phi_T^{w,z}(x, y) = T(w + x, z + y) - T(w + x, y) - T(x, z + y) + T(x, y).$$

As in Mandal's Lemma 2.1, $\phi_T^{w,z} \in \ell_\infty(G, E)$ and

$$\|\phi_T^{w,z}\|_\infty \leq \text{BLip}(T) \|w\| \|z\|.$$

Define

$$P(T)(w, z) = M(\phi_T^{w,z}).$$

The additivity identities for the functions $\phi_T^{w,z}$ are exactly those used in [1, Theorem 2.3]; translation invariance and linearity of M imply that $P(T)$ is additive in each variable. The preceding bound gives continuity, so the standard rational-scalar argument makes $P(T)$ bilinear. Moreover

$$\|P(T)(w, z)\| \leq C \text{BLip}(T) \|w\| \|z\|,$$

hence $\|P\| \leq C$.

If T is already bilinear, then $\phi_T^{w,z}$ is the constant function with value $T(w, z)$, so $P(T)(w, z) = M(\widehat{T(w, z)}) = T(w, z)$. Therefore P is a bounded linear projection from $\text{BLip}_0(X, Y; E)$ onto $\text{Blin}(X, Y; E)$. $\square$

Corollary 1. *Question 1 has an affirmative answer whenever E is complemented in its bidual. Equivalently, by Lindenstrauss' characterization quoted in [2], this includes Banach spaces isomorphic to complemented subspaces of dual Banach spaces.*

Theorem 2 (split barycenter maps). *Let X, Y be Banach spaces. Suppose the barycenter maps $\beta_X : \mathcal{F}(X) \rightarrow X$ and $\beta_Y : \mathcal{F}(Y) \rightarrow Y$ admit bounded linear right inverses $s_X : X \rightarrow \mathcal{F}(X)$ and $s_Y : Y \rightarrow \mathcal{F}(Y)$. Then for every Banach space E , the subspace $\text{Blin}(X, Y; E)$ is complemented in $\text{BLip}_0(X, Y; E)$. The projection can be chosen with norm at most $\|s_X\| \|s_Y\|$.*

Proof. Let $Z = \mathcal{F}(X) \widehat{\otimes}_\pi \mathcal{F}(Y)$ and $W = X \widehat{\otimes}_\pi Y$. The two-Lipschitz linearization identifies $\text{BLip}_0(X, Y; E)$ isometrically with $L(Z, E)$.

Let

$$q = \beta_X \otimes \beta_Y : Z \rightarrow W, \quad s = s_X \otimes s_Y : W \rightarrow Z.$$

Then $qs = I_W$, and $\|s\| \leq \|s_X\| \|s_Y\|$. Hence

$$\pi = sq : Z \rightarrow Z$$

is a bounded projection onto $s(W)$.

Under the linearization, a two-Lipschitz map is bilinear precisely when its linearization factors through q ; this is the content of Mandal's Proposition 4.3, equivalently the vanishing on $\ker \beta_X \widehat{\otimes}_\pi \mathcal{F}(Y) + \mathcal{F}(X) \widehat{\otimes}_\pi \ker \beta_Y$. Define, for $A \in L(Z, E)$,

$$\Pi(A) = A\pi = Asq.$$

Then $\Pi(A)$ factors through q , so it corresponds to a bilinear map. If $A = Bq$ already factors through q , then

$$\Pi(A) = Bqsq = Bq = A.$$

Thus Π is a projection onto the bilinear subspace and $\|\Pi\| \leq \|s\| \leq \|s_X\| \|s_Y\|$. $\square$

Remark 1. *The second theorem gives an all-codomain positive answer for every pair (X, Y) whose free-space barycenter quotients split linearly. In particular, finite-dimensional X and Y satisfy this hypothesis by choosing a basis and lifting basis vectors to their Dirac masses. The theorem is not claimed to cover arbitrary Banach spaces.*

Limitations and Novelty Check

These results do not settle Mandal’s unrestricted question. They leave open the case where E is not complemented in E^{**} and the free-space barycenter maps for the domains do not split.

Local index searches for arXiv:2512.05640, “two-Lipschitz”, “BLip_0”, “Blin”, and “vector-valued invariant mean” found no existing run packet for this paper. Web searches for the exact title, for “BLip_0(X,Y;E)”, and for “Blin(X,Y;E)” found the source paper and the one-variable predecessor arXiv:2506.09324, but no later complete answer. The bidual-complemented codomain result relies on known invariant-mean theory, especially Kania [2]; the present contribution is the direct application to Mandal’s Question 1 and the split-domain operator reformulation above.

References

References

- [1] A. Mandal, *Projection from space of two-Lipschitz operators onto the space of Bilinear maps*, arXiv:2512.05640, 2025.
- [2] T. Kania, *Vector-valued invariant means revisited once again*, arXiv:1607.03387, 2016.
- [3] G. Godefroy and N. J. Kalton, *Lipschitz-free Banach spaces*, Studia Math. 159 (2003), 121–141.